

SECTION A (25 marks)

1. Evaluate the following:

a) $\lim_{x \rightarrow -\infty} \frac{2x^2+3}{-x^3+x}$ [3 marks]

b) $\lim_{x \rightarrow \infty} \sqrt{x^2 - 1} - \sqrt{x^2 + 1}$ [3 marks]

c) $\lim_{x \rightarrow 2^+} \frac{10-3x-x^2}{|2-x|}$ [4 marks]

2. a) Find the gradient of $\ln(x + y) = e^{xy}$, when $x = 0$. Leave your solution in the simplest exact value. [4 marks]

b) Given

$$f(x) = \begin{cases} x^2 - 3, & x < 2 \\ \sqrt{x-1}, & x \geq 2 \end{cases}$$

Determine whether $f(x)$ is differentiable at $x = 2$. Justify your solution. [5 marks]

3. Find the critical values of $f(x) = \frac{1}{4}x^4 - 2x^2 + 4$ and specify the nature of the following values. Then, find the values of x where change of curvature happens. Hence, state the region where f is concave upwards and downwards. [6 marks]

SECTION B (75 marks)

1. Given $z = x + iy$ where x and y are real numbers. Express $\frac{z+i}{zi+1+i} = 2$

as two equations in terms of x and y . Hence, evaluate x and y . [5 marks]

2. a) Solve $\log_4(x+4) - 5\log_{(x+4)} 16 = 4$ by using substitution method. Leave your solution(s) in exact form. [5 marks]

- b) Solve:

$$\frac{|x^2 - 1|}{3x^3 - x^2 - 8x - 4} \leq 0$$
 [5 marks]

3. a) A function is a specific type of relation where each input has exactly only one output. State the two types of function relations. [2 marks]

- b) The function f and g are defined as follows:

$$f: x \rightarrow x^3, \quad x \in \mathbb{R}$$

$$g: x \rightarrow 2^{x-3}, \quad x \in \mathbb{R}$$

Find $(f \circ g)(x)$. Hence, $(g^{-1} \circ f^{-1})(x)$ in the simplest form. [6 marks]

4. Given the function $m(x) = 3e^{-x} - 4$

- a) State the domain and range of $m(x)$. [2 marks]

- b) Prove whether $m^{-1}(x)$ exists. [3 marks]

- c) Determine $m^{-1}(x)$ and sketch the graph $m(x)$ and $m^{-1}(x)$ on the same axis. Label your graph completely. [6 marks]
5. a) Determine the asymptotes for the function $f(x) = \frac{x^2-4}{9-x}$.
Then, state the region where the function $f(x)$ is continuous. [6 marks]
- b) Given the function
- $$h(x) = \begin{cases} x^2 - 4x + 3k + 10 & , \quad x \leq 0 \\ 2e^x - 5k & , \quad 0 < x < 3 \\ \frac{2x-1}{x+2} & , \quad x \geq 3 \end{cases}$$
- i. Find $\lim_{x \rightarrow 0^-} h(x)$ and $\lim_{x \rightarrow 0^+} h(x)$. Hence determine the value of k such that $h(x)$ is continuous at $x = 0$. [5 marks]
- ii. Determine whether $h(x)$ is continuous at $x = 3$. State your reason. [3 marks]
6. a) Given that $x = \frac{1}{\sqrt{1+t^2}}$ and $y = \frac{\sqrt{1+t^2}}{t}$ where t is a non-zero parameter.
- i. Show that $\frac{dy}{dx} = \frac{1+t^2}{t^3}$ [6 marks]
- ii. Find $\frac{d^2y}{dx^2}$ when $t = 1$ [6 marks]
- b) If $y = \sin(x^2 + 1)$, show that [5 marks]

$$x \frac{d^2y}{dx^2} - \frac{dy}{dx} + 4x^3y = 0$$

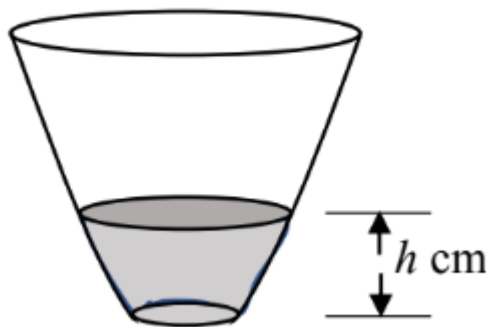
7. a) The function f is defined by

$$f(x) = (x - 2)(3 + 2x)^2$$

- i. Find the y –intercept. [1 mark]
- ii. By using the first derivative test $f'(x)$, find the intervals of x where f is increasing and decreasing. [5 marks]

- b) A vase with a circular cross-section is shown in the figure below.

Water is poured into the vase. When the depth of the water is h cm, the volume of the water is given by $V = 4\pi h(h + 4)$, $0 \leq h \leq 25$. Water is poured into the vase at a constant rate of $80\pi \text{ cm}^3 \text{ s}^{-1}$. Find the rate of change of the depth of the water when $h = 6$.



[5 marks]